

Project Assessment (Handout for Learner)

Module 6: Java Programming for Developers

Approved Assessment Centre: BELLS Institute of Higher Learning Pte Ltd
Candidate Name:

Copyright 2023 © Trent Global Pte Ltd & BELLS Institute of Higher Learning Pte Ltd.
All Rights Reserved

VERSION CONTROL RECORD

Version	Effective Date	Changes	Author
1.0	29th Aug 2023	First Edition	Program Creator: Chor Kun Xin Course Developer: Poh Soon Hock

Copyright 2023 © Trent Global Pte Ltd & BELLS Institute of Higher Learning Pte Ltd.
All Rights Reserved

Candidate's Name (as in NRIC):	
Candidate's NRIC (Please use last four figures of NRIC, e.g. "SXXXX123A"):	S/T XXXX _ _ _ / _
Assessor's Name:	
Assessor's NRIC: (Please use last four figures of NRIC, e.g. "SXXXX123A"):	S/T XXXX _ _ _ / _
Duration of Assessment:	
Allocated Duration:	15 minutes
Start Time	
End Time	

Project Assignment

This **Project Assignment with Oral Clarification** is to gather evidence for the following learning Outcomes:

- **LO1:** Determine when use cases would require the use of Java.
- **LO4:** Design simple classes with single-level inheritance, overridden methods, interfaces, and abstract classes.
- **LO5:** Design and interpret UML class diagrams.

Copyright 2023 © Trent Global Pte Ltd & BELLS Institute of Higher Learning Pte Ltd.
All Rights Reserved

Portfolio

Java Programming for Developers

INSTRUCTIONS FOR CANDIDATE

Requirements

In this portfolio project, you will create a **software design that uses Java** to solve a particular problem. You should choose a problem that justifies the use of Java. The software design must meet the following requirements

- The use case should make use of the strengths of Java
- The software design must consist of the following:
    - A write-up that describes the problem, why Java is suitable and why alternatives to Java will not be suitable
    - A class diagram that shows all the classes and their relationships with each other. The class diagram should show at least **two** attributes relevant to the business case.
    - An object diagram that shows one possible state of all the classes.
- The software design must be submitted as one PDF document.

Marking Criteria

Write-Up

The software design report must have a write-up that covers the following points:

- One justification for using Java – your justification should mention a strength of Java
- State one possible alternative instead of using Java, and why it is not suitable

Class Diagram

Copyright 2023 © Trent Global Pte Ltd & BELLS Institute of Higher Learning Pte Ltd.
All Rights Reserved

- Your class diagram must consist of the following:
    - At least three classes
    - You must have at least **one** class that can be instantiated.
    - You must have at least one directed association
    - You must demonstrate **inheritance** at least once
    - You must override at least one **method**
    - You must have at least one **interface** or **abstract base class.**

Object Diagram

- Your object diagram must consist of the following:
    - For each class, show at the values for at least one object

Submission

- The software design report must consist of the write-up, the class diagram and the object diagram in **one** PDF file.
- The PDF file must be submitted in the LMS

Instructions to Candidates for Oral Clarification

The candidate will be required to present the answer of project Assignment.

During the Presentation, learner is able to demonstrate the ability for the following Learning Outcomes:

- **LO1:** Determine when use cases would require the use of Java.
- **LO4:** Design simple classes with single-level inheritance, overridden methods, interfaces, and abstract classes.
- **LO5:** Design and interpret UML class diagrams.

Individual

This project work is an individual assessment

Copyright 2023 © Trent Global Pte Ltd & BELLS Institute of Higher Learning Pte Ltd.
All Rights Reserved